

Experiment 'p_4_8_seperate_lines' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_8_seperate_lines

Natural language statement: Write a method that reads a word and returns each character of the word on a separate line.

Method signature: p_4_8_seperate_lines(s: string) returns (chars: string)

Ensures

- `|chars| == 2 * |s|`
- `forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i] == '\n'`
- `forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that reads a word and returns each character of the word on a separate
↪ line.

The signature should be:

p_4_8_seperate_lines(s: string) returns (chars: string)

The method should respect the following contract:

ensures |chars| == 2 * |s|, ensures forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i]
↪ == '\n', ensures forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dmfe4pOhcvoskJsQ5OxGkNsEwIfr

Model: gpt-5.5-2026-04-23

Created at: 1780492200

Finish reason: stop

Usage: 394 tokens in, and 2132 tokens out

```
{
  chars := seq(2 * |s|, i requires 0 <= i < 2 * |s| => if i % 2 == 0 then s[i / 2] else
    ↪ '\n');
}
```

Final Program

```
method p_4_8_seperate_lines(s: string) returns (chars: string)
  ensures |chars| == 2 * |s|
  ensures forall i :: 0 <= i < |s| ==> i % 2 == 1 ==> chars[i] == '\n'
  ensures forall i :: 0 <= i < |s| ==> i % 2 == 0 ==> chars[i] == s[i / 2]
{
  chars := seq(2 * |s|, i requires 0 <= i < 2 * |s| => if i % 2 == 0 then s[i / 2] else
    ↪ '\n');
}
```

Total Token Usage

Input tokens: 394

Output tokens: 2132

Reasoning tokens: 2047

Sum of 'total tokens': 2526

Experiment Timings

Overall Experiment started at 1780492199763, ended at 1780492243934, lasting 44171ms (44.17 seconds)

Iteration #1 started at 1780492199763, ended at 1780492243934, lasting 44171ms (44.17 seconds)